

Mechanical Engineering Senior Design Project

(ME SDP)

ME SDP 115: Smart Micro-Particle Manipulation Chamber

Project Report

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Executive Summary

The goal of this design project is to create a device that allows the study and manipulation of microfluidic particles under electromagnetic fields at a macroscopic level. This is, in large part, a tool to support Dr. Kaiyan Yu's research on microfluidic particles and their reaction to electromagnetic stimuli. A device like this is very useful for both teaching students these concepts and demonstrating her research. After conducting thorough research, an effective design that meets these requirements and other limitations was developed.

This project is a continuation of last year's ME SDP304 with the same project title. While the previous team succeeded in delivering a prototype, it failed to meet a few requirements. Namely, the ferrofluid was never able to rise above the first row of magnets. As such, this year's goal is to improve upon that design and ensure that all of the requirements, including functionality, are met.

This design consists of a few main components: the design housing, a 9x9 array of electromagnets, a storage array for the electromagnets, a tank containing ferrofluid suspended in a 25% concentration saltwater medium, a tablet which features a custom UI application to control the electromagnets, and other necessary electrical components. The design housing and electromagnet array are 3D printed from PLA, while all other components are purchased from assorted vendors. The design housing includes some features important for usability and safety, such as ventilation slots to let the electrical components cool properly and a slight downward slope and exit channel to ensure that any leaks in the tank will not cause damage to the electrical components.

The UI application running on the external tablet will allow the user to control which electromagnets are turned on and off. A representative 9x9 array will be shown on the device, which the user can draw shapes on to create those same shapes with the ferrofluid in the tank. A small camera is also pointed at the tank, such that the image created by the ferrofluid is represented on the UI application as well.

The system will be powered with a 12V, 29A DC power supply. For safety reasons, it was ensured that the current draw of the system will never exceed 16.2A, maintaining a sizeable safety margin. This power supply will primarily be used to power the 81 12V electromagnets used to modulate the ferrofluid. They need to have enough effective lifting force to pull a significant amount of ferrofluid from the bottom of the tank and lift it up to any point in the array, using other magnets along the way to assist with the lifting.

The budget for this project was proposed to be \$1500, but was reduced to \$750. After a successful budget increase proposal, it is now \$950. A vast majority of that budget is spent on strong electromagnets to ensure that the ferrofluid can rise above the first layer. A large portion of the rest of the budget, around \$170, is reserved for required electrical components and a new microcontroller, while about \$50 is required for a new tank to meet the new requirements for this year's design.

1. Problem Statement

1.1 Problem Scope

The primary objective of this project is to create a chamber that utilizes magnetic and electric fields to study and manipulate microparticles. Oil-based ferrofluid is utilized as the micro-particle stored within a salt-brine medium, while an array of electromagnets is modulated with a power supply to manipulate the ferrofluid. The electromagnets are controlled with an external tablet utilizing a custom-made application, allowing a user to modulate the state of each individual electromagnet with a custom UI.

This project is a continuation of ME SDP 304 from last year with the same title. The previous year successfully completed a design prototype but encountered some issues with experimentation and usage. The previous team's 7x7 array of electromagnets was not able to raise the ferrofluid above the first row of electromagnets. The project this year focuses largely on improving the ideas created last year and developing an improved, functioning device following the same principles.

This device will be utilized primarily as a learning and demonstration tool to show how microparticles react when they experience an electric or magnetic field. By creating a device at a larger scale, the behavior of particles can be seen at the macroscopic level and therefore, allows for easier learning and understanding of complex topics. Additionally, Dr. Kaiyan Yu will utilize the Smart Micro-Particle Manipulation Chamber for her research on microfluidic particles.

1.2 Technical Review

Electromagnetic manipulation of microparticles has practical applications across many fields, such as nanotechnology, biomedical engineering, and materials science. It is founded upon the principles of electricity and magnetism; specifically, the way that electric charges and current can induce electric and magnetic fields and how these fields can be shaped to have noticeable physical effects on small particles. By learning how to design field patterns, systems can be constructed that can position and move microparticles with precision, allowing for a wide range of possible tools and accomplishments utilizing the potential of the technology.

Electromagnet manipulation systems evolved over the years from static electromagnet arrays that were only capable of producing basic fields and mainly linear motion to dynamic systems where fields that vary in strength and direction over time can be adapted. This more nuanced modulation of variables within the system is greatly utilized within this project to allow for precise microparticle movement.

For this project, ferrofluid is utilized as the object of manipulation. Ferrofluid consists of iron-based magnetic nanoparticles suspended in a carrier fluid, typically a polar oil-based solution, and has noticeable physical effects when exposed to a magnetic field. The particles align along field lines and can be seen forming characteristic surface "spikes." When the field strength and distribution are changed, the structure can be strengthened and moved, proving that ferrofluid is a great medium for demonstrating magnetically driven behavior.

In order for this project to be successful, the ferrofluid suspension must remain clear and visible at all times. This is accomplished in two different ways. Firstly, the ferrofluid must be suspended in an

appropriate medium. If a polar, oil-based ferrofluid is utilized, then an appropriate nonpolar suspension solution would be a good option. Secondly, a clear container for the solution is required, and it must be ensured that the container does not stain in any way due to a reaction with the ferrofluid or suspension fluid. Glass is an appropriate material for this selection...

The Smart Micro-Particle Manipulation Chamber is designed to help people understand how magnets change and manipulate electromagnetic fields. The glass chamber is easily seen through, and the electromagnets in the back are fully responsive to an intuitive touch-based tablet system. This allows people of all ages to easily manipulate the fluid inside the tank and observe the magnetic field in a simplistic manner.

2. Project Requirements

[MESDP-015-MPMC-01] The Smart Micro-Particle Manipulation Chamber is designed with a glass housing unit mounted in the center, allowing for the containment of fluid. The clear glass allows the observer to see the magnetic fields.

[MESDP-015-MPMC-02] The system, in combination with the mechanical, electrical, and computer components, allows the user to control and understand the relations of a small magnetic field.

[MESDP-015-MPMC-04] Various electrical components, including the use of computers, allow the electromagnets to be controlled with a range of parameters.

[MESDP-015-MPMC-05] The use of ventilation throughout the housing unit of the chamber allows for the regulation of temperature and a controlled environment for experimentation.

[MESDP-015-MPMC-08] The utilization of a tablet, connected to the system, allows for a friendly and intuitive way to remotely control and monitor the system as the experiment progresses.

[MESDP-015-MPMC-11] The Smart Micro-Particle Manipulation Chamber is designed with all safety regulations and standards in mind, and has been thoroughly checked by not only our group, but others as well.

[MESDP-015-MPMC-12] The tablet allows for an extremely user-friendly interface, which could be easily accessible for people of all ages, including children.

[MESDP-015-MPMC-13] The housing is built with maintenance in mind, as instead of one complete unit, it is instead made up of individual parts, connected together post-processing.

[MESDP-015-MPMC-18] There are several bills of materials regarding all the information surrounding the different parts used in the creation of this system, allowing for easy replacement.

[MESDP-015-MPMC-19] The Smart Micro-Particle Manipulation Chamber conforms to the budget within reason, and any extra money the group may need is being asked for through methods given to us by our advisors.

3. Design Description

3.1 Overview

The Smart Micro-Particle Manipulation Chamber is designed to demonstrate the effect of magnetic fields on fluid particles. Once completed, Dr. Yu will utilize the device to conduct

demonstrations and further her research on microparticles and microfluidic devices on a macroscopic scale. An oil-based ferrofluid solution will serve as the magnetic microfluidic particle for the device, and will be submerged in a saltwater medium. The device will consist of a few major parts: the housing, the tank, the electromagnets, the electromagnet holder, the electrical components, the UI interface tablet, and the microcontroller.

The device will be controlled by an external tablet operating a custom UI application. By drawing a shape on the application, those corresponding magnets will be enabled and the shape will be created by the ferrofluid inside the device. An installed camera will constantly be viewing the status of the ferrofluid tank and will also be displayed on the UI of the tablet.

Since this project was also a previous year's SDP, the main goal was to improve on the previous design and create a working prototype rather than to fully innovate. As such, many of the design choices were intended to directly solve flaws that were noted in the previous design. The subsections below for each individual design component will highlight the decisions that were made to improve the design for each component.

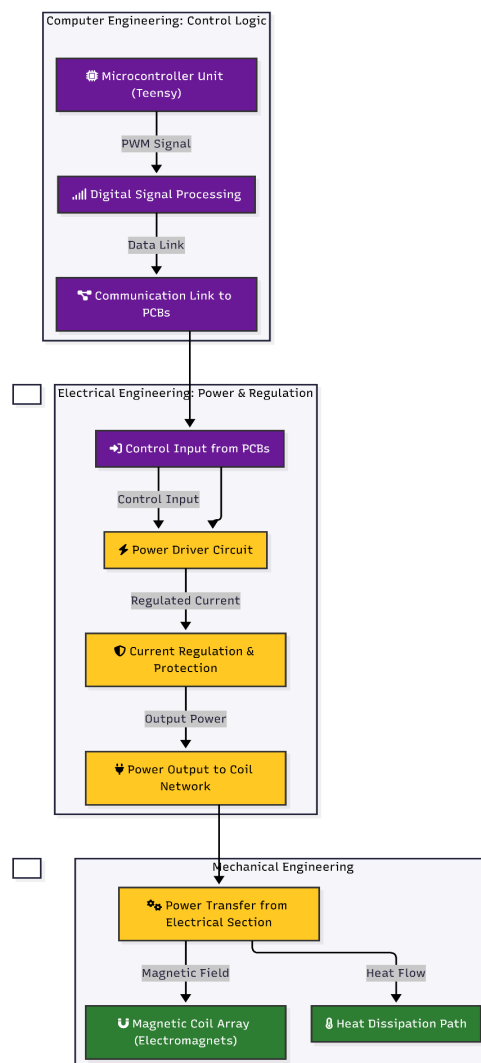


Figure 1: System Level Block Diagram for the Smart Micro-Particle Manipulation Chamber

3.2 Detailed Description

3.2.1 Device Housing

The housing for the device is a 285mm x 285mm x 320mm hollow rectangular prism structure made of 3D printed PLA. This structure is printed in 8 parts: the base, different left and right walls, a front wall, a back wall, the lid, a front mounted tablet holder, and an internal magnet holder. All of the walls are screwed into the base using countersunk M6 40mm screws. The left and right walls are additionally attached to the front wall using M6 40mm screws to improve the rigidity of the overall structure. The back wall slides into place guided by slots in the left and right walls, and is designed for easy removal in the event that any electrical components need to be adjusted. The lid slides into place using cut-outs rails at the top of the left and right wall.

The front wall has a cut-out equal to the size of the tank with grooves on the sides, such that the ferrofluid tank can slide in and be held securely in place. The magnet holder rests neatly behind the tank in a secure position, such that the magnets are in direct contact with the tank. This ensures that the strongest electromagnetic force possible can be applied to the ferrofluid solution in order to produce more visually interesting results.

The housing for last year's design was printed in three parts: a sliding back panel, a sliding top panel, and then the rest of it as one large rectangle. This design was problematic for a few reasons, but namely its makeability. The print took multiple days, and the design was hard to modulate/upgrade if it became necessary in the future. By splitting this housing into multiple smaller pieces that assemble together later, these problems were alleviated and the device was made significantly more manufactureable.

3.2.2 Tank

The tank used in the design is constructed of 1.4 mm thick glass, with outer dimensions of 8" x 8" x 0.5". The tank medium is a saltwater brine mixture at approximately 25% concentration. In order to ensure that the ferrofluid does not react with anything in the tank and cause discoloration of the glass, a non-reactive sealant material was used. For this design in particular, epoxy was used to seal the tank inside and outside on all the seams, both in order to prevent any fluid leakage and prevent reaction between the sealant and oil-based ferrofluid.

Before the tank was filled with the fluid medium, it was cleaned thoroughly with a cleaning agent and then distilled water. Afterwards, the required mass of salt was added, followed by distilled water until the desired volume was reached. The medium was then left for approximately two weeks so that the saltwater brine can come to an equilibrium and fill any impurities in the glass, reducing the chance of a reaction between the ferrofluid and the tank enclosure.

Internal testing prior to the prototype revealed that the tank and solution assembly was one of the more difficult aspects of assembly. Multiple factors caused this: internal manufacturing of the tank, varying types of available glass used, solution used, etc. Utilizing last year's research as a baseline, the team focused on using a saltwater brine as the suspension solution, as that was the previous solution with the most visibility. Varying concentrations of saltwater were tested, and it was determined that the highest possible concentration at room temperature led to the clearest solution.

The primary flaw with last year's design was the inability of the device to lift the ferrofluid above the first row of magnets. The prior design used a 7x7 array of larger (25mm diameter, 25mm height) 12V magnets with a similarly sized tank. When the magnets were powered, the ferrofluid could only be lifted to the height of the first row of magnets, and was never able to travel any higher than that without external stimulus. To solve this problem, the internal tank thickness was more than halved and the glass thickness was reduced from approximately 6.4 mm to 1.4 mm.

3.2.3 Electromagnets

The JSP-1515 is a compact, round DC holding electromagnet widely utilized in precision robotics and automated material handling due to its balance of small form factor and reliable suction. Measuring 15mm both diameter and height, this component operates on a 12V, consuming approximately 1.4W of power while generating a holding force of roughly 20N. The JSP-1515 was preferred over the magnets utilized in the previous iterations of the Smart Micro Particle Manipulation project due to its small size. Smaller magnets allowed us to create a larger electromagnet array without the need to design a massive enclosure and housing for the magnets and electronics.

3.2.4 Tablet

The Byybuo A10 is a 10.1 inch touchscreen tablet with wifi capabilities. This allows it to be responsible for displaying the user interface and receiving touch inputs for users to interact with our web application. This app requires wifi connection to our LAN connecting the tablet to a Raspberry Pi 5 running the server. The tablet provided a responsive and intuitive interface during testing enabling real-time interaction with the system.

3.2.5 User Interface Application

The User Interface (UI) application was implemented as a web-based application hosted on the Raspberry Pi, with the tablet serving as a client connected over a local wireless network. The interface displays a 9x9 grid of buttons, with each directly corresponding to an electromagnet in the physical array. Users can select which electromagnets to toggle directly through the tablet to create their desired pattern using the ferrofluid. Data is sent to the Raspberry Pi, which processes the input and forwards it to the Teensy 4.1 microcontroller, subsequently controlling the electromagnets using PWM signals.

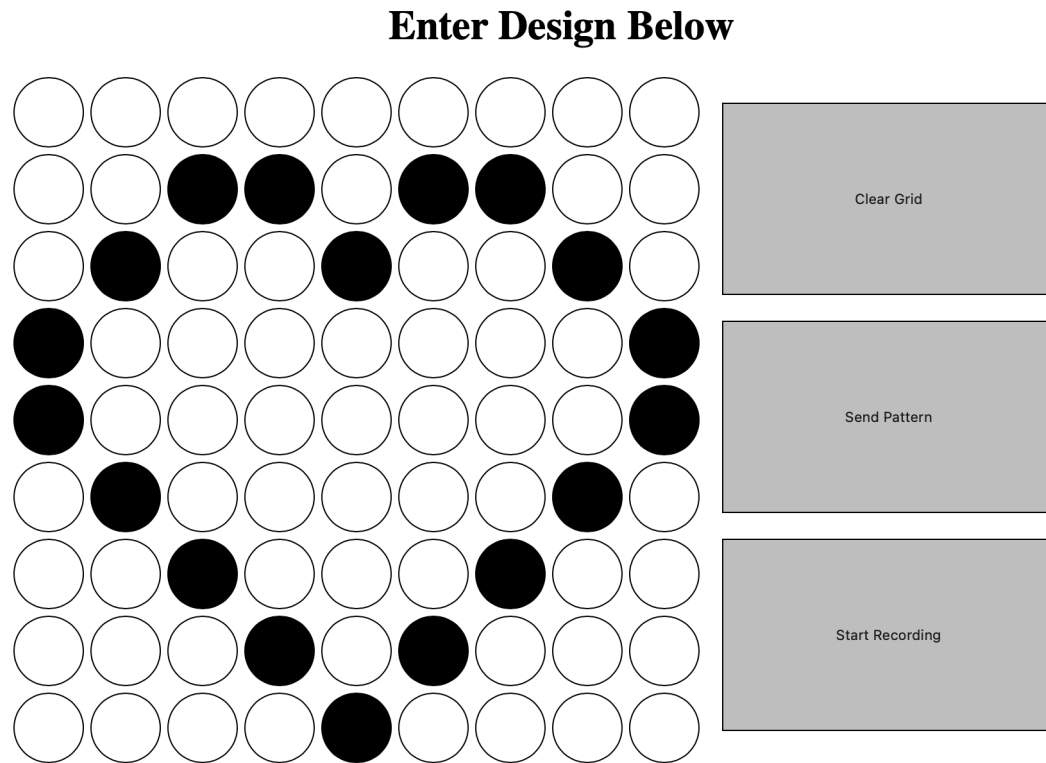


Figure 2: Tablet User Interface and Pattern Generation Grid

3.2.6 Electromagnet Control System

It was necessary to control the magnetic field strength of each individual electromagnet. This was accomplished by designing a PCB whose tasks are to power and control each magnet. The reason this is necessary is because gravity will pull the ferrofluid down towards the bottom magnet if several pixels are ignited in the same column. The simplest way to control the magnetic field, without having to deal with any additional complex adjustable current circuitry, is by pulse-width modulating (PWM) the supply current that is fed to the electromagnets. The following requirements were set for the electromagnet control PCBs:

- One single board to control each row of electromagnets to simplify the code (9 PCBs in total)
- Thin and wide form factor so it would fit behind the electromagnet matrix and all the wires would reach their dedicated connector.
- An IC that could handle PWM using a serial protocol
- An alternating pattern of electromagnet polarities designed into the board

The PCA9685 proved to be able to provide decent PWM capability at the same time as providing a totem-poled (logic high) signal on an array of outputs. The PCBs link the pulse-width modulated outputs of the PCA9685 to the inputs of a ULN2803 Darlington transistor array.

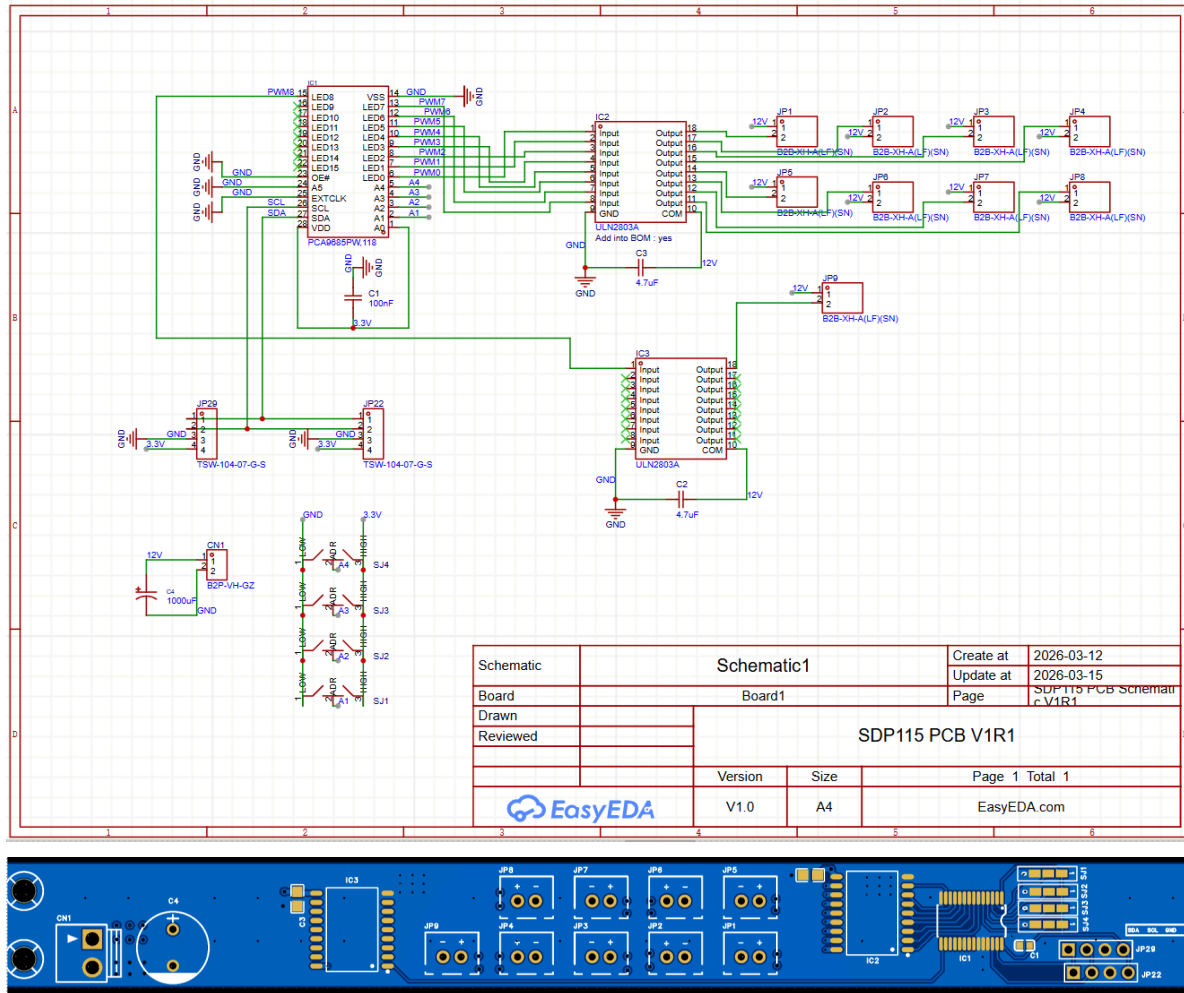


Figure 3: Electromagnet Control PCB Schematic and Board Layout

Each PCB needed to have a unique address, which was accomplished by using 4 solder jumpers in the form of exposed copper pads. To assign an address, solder is added to the middle pad of the solder jumper and then bridges either to 1 or 0. 0 connects to GND and 1 connects to either the A1, A2 and A3, and A4 (depending on which solder jumper is being used) pin of the LED driver. Since the Teensy needed to communicate to 9 PCBs 4 bits were needed in order to assign 9 addresses. The addresses are set to 0x41, 0x42, 0x43, ..., 0x49. Each address corresponds to its respective PCB.

3.2.7 Power Supply System

The system uses power from the wall and an AC to DC power supply as the main source power. A MEAN WELL LRS-350-12 was chosen as the power supply because it supplies 12V and 29A, leaving enough current for all of the system components, which were measured to consume a maximum of 16.2A. One design challenge that we encountered was that we needed the power supply to supply 12V to 9 PCBs, however, the power supply only had 3 terminals. A more sophisticated power distribution strategy was necessary. The solution we found was to design a power distribution PCB to supply 12V. Each PCB

powers 9 magnets, each magnet consumes 200mA at full hold, and with 9 PCBs in total, the power distribution PCB needed to be able to distribute more than 16.2A. A large polygon copper pour assigned to 12V was added to the PCB to safely handle this load. The JST-VH connector was used to connect the power distribution PCB to the electromagnet control PCB. This connector is typically rated for up to 10A, which provides a massive safety margin for the 1.8A per PCB requirement.

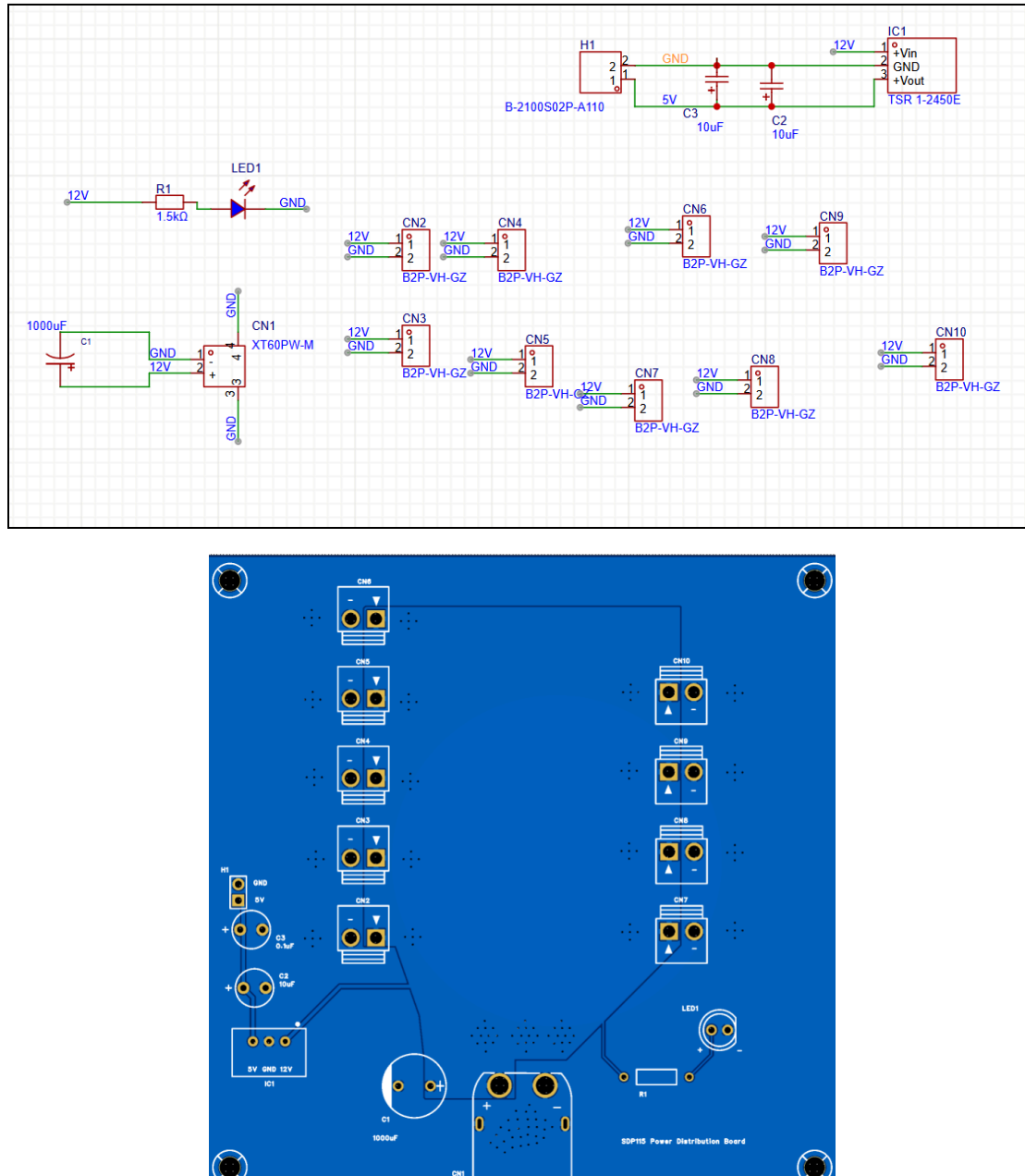


Figure 4: Power Distribution PCB Schematic and Board Layout

A TSR 1-2450 DC DC converter was also added to the power distribution PCB to provide a 5V source to the Teensy and Raspberry Pi.

3.2.8 Microcontroller

The Teensy 4.1 is the microcontroller used to handle the majority of the electromagnet control. Using the SDA and SCL outputs of the Teensy to communicate with the PCA9685 LED drivers, enabling independent control of the system's 81 electromagnets with pulse-width modulation. In testing, this allowed for the use of animations to input user created patterns moving the ferrofluid vertically and horizontally.

3.3 Use

The Smart Micro-Particle Manipulation Chamber is easily user operated by using the attached tablet that features a simple graphical user interface, in which the user is able to interactively control the electromagnet array, without physically touching the device. The interface is on a tablet, that is mountable on a focal point (Raspberry Pi 5) and is connected between the user and hardware. By connecting them to the control panel, it is possible to turn off and on individual electromagnets of the 9x9 array, control the strength of the magnetic fields using PWM control and design their own patterns to move ferrofluids in the tank.

System backend is written in Fast API, and all the communication between the user interface and all of the hardware components, such as Teensy 4.1 microcontroller and PCB control boards (using PCA9685) are performed. The user account data is then saved into a PostgreSQL database that is a secure and scalable method of storing and retrieving user profiles and session data.

A combination of these hardware and software elements forms a complete and single platform to precisely manipulate microparticles. Its remote operation capability, ability to control the individual electromagnets, and a touch-based interface accessible interface make the chamber highly adaptable to both structured research and educational demonstrations, and do not need any specific technical expertise to work.

3.4 User Manual

To power on the device, plug the wall-mount cable into the side of the housing and flip the ON switch. Allow approximately one minute for the Raspberry Pi to boot before proceeding. On the tablet, connect to the system's dedicated Wi-Fi network, then open Google Chrome on the table to access the Pi's IP, which will load the control web application.

The interface displays a 9x9 grid of buttons, each corresponding to one electromagnet in the physical array. Tap cells to select a pattern, then press the "Send pattern" button to animate the ferrofluid to the corresponding electromagnets and produce the pattern in the ferrofluid tank. To enter a new pattern press the "Clear grid" button and repeat the prior process to enter a new pattern. A Toggle Camera button is also present on the interface, intended to display a live feed of the tank, though this feature is not yet functional in the current prototype. Note that the exact button names and UI layout are subject to change, engineers should verify these against the final implementation.

Before submitting a new pattern, wait for the ferrofluid sediment to fully settle from the previous activation. Submitting patterns too quickly can cause glass staining and unclear results. Keep the device in a cool, well-ventilated area during use, as there is no internal fan. Do not operate the device near warm or

hot objects. Be aware that some electromagnets may not reach desired strength levels due to minor hardware faults identified during testing.

To power off, press the OFF switch, it is safe to cut power directly without a software shutdown. If an electromagnet is unresponsive, first check whether the JST connection has come loose; if the issue persists, replace it with one of the spare magnets included with the device. If the ferrofluid appears stuck or stained, the tank can be removed from the front wall slot, cleaned with soap and water, and refilled with a fresh 25–28% saltwater brine solution and approximately 10mL of ferrofluid. Allow the refilled tank to condition for two weeks before use to prevent glass staining.

4. Implementation

The working prototype exists to address Dr. Kaiyan Yu's need for a macroscopic scale tool utilizing electromagnetic fields in order to simulate microfluidic particle movement. This device should be both simple and useful, in other words, it must be appropriate for both high level research and educational purposes for those who know nothing about electromagnetic forces. Since this project is a continuation of a previous SDP, a majority of the working time was spent improving any problems present in the previous design and ensuring that functionality and simplicity were expressed in the final product. The list at the bottom of this section contains the problems and subsequent solutions that were addressed with this prototype.

The SMPMC prototype device accomplishes these needs by combining a clear, stain-free ferrofluid tank environment with precise electromagnetic manipulation. The ferrofluid is suspended in a highly concentrated saltwater solution, contained within a glass tank permanently sealed with epoxy. All of these variables were independently tested to prove that there would be no staining, discoloration, or other long-term effects on the visibility of the solution. A piece of paper serves as a white background placed behind the tank to create visual clarity for the ESP32 camera.

Directly behind the white background is the magnet holder, keeping the 9x9 array of electromagnets evenly ordered. A 1mm gap and open backing allows for proper airflow to ensure that no components in contact with the magnets reach dangerous temperatures. The electromagnets are connected to and controlled by 10 PCBS, 1 for each row of magnets and one final larger PCB specifically for power distribution. A front mounted 3D print addition contains both the UI control tablet and the ESP32 camera. Any user can input their desired shape or pattern on the tablet, sending a signal to the electrical components and, after a short delay, creating the desired pattern inside of the ferrofluid enclosure. The ESP32 camera records the resulting ferrofluid movement and final pattern, allowing for further behind the scenes data analysis.

All of these components are stored within a custom designed and 3D printed PLA housing. This housing is divided into 7 parts in order to achieve good manufacturability and ease of access to the internal electrical components. The tank is stored within a slot in the front wall, and all electrical components are installed on the walls using mounted screw holes. The back wall is specifically designed to be easily lifted out and laid flat on a table, allowing any electrical edits to be made easily.

Problem: Ferrofluid could not be raised above the first layer of magnets.

Solution: 1. The 7x7 array of 25mm magnets was replaced with a stronger 9x9 array of 15mm magnets.
2. The glass tank was halved in diameter in order to ensure that the ferrofluid was closer at all times to the magnets.

Problem: The design was designed as one large 3D printed part, causing manufacturing issues

Solution: The housing was remodeled to consist of 6 main parts which were connected with joints such as the dovetail and M6 40mm screws in order to improve 3D printing manufacturability.

Problem: Accessing the internal electrical components becomes difficult when the device is fully constructed.

Solution: The modular housing allows for the back wall (or any other individual wall) to be easily removed, meaning that the internal components can be accessed from any one individual side without disassembling the entire device.

Procedure Followed for Filling Tank:

1. The inside and outside of the tank was cleaned with a generic glass cleaner and rinsed with distilled water.
2. In a separate container, one part kosher salt and two parts water. This solution was then mixed thoroughly in order to achieve a very highly concentrated saltwater solution (around 28%). More salt was added than necessary, since the goal was achieving the highest possible concentration rather than a specific value.
3. The saltwater solution was then moved into the tank and was allowed to sit undisturbed for two weeks. This procedure was particularly important in order to prevent any staining of the glass enclosure. The saltwater solution was given time to coat any impurities present in the glass, ensuring that the ferrofluid would not react and reduce visibility.
 - a. Previous testing proved that the ferrofluid would not react with either the glass or the saltwater solution, but impurities in the glass would react with the ferrofluid and lead to visibility issues.
4. After two weeks had passed, a pipette was used to pull some ferrofluid directly from the container and place it at the bottom of the tank. Approximately 10mL of ferrofluid was added to the tank (enough to form a complete layer on the bottom of the tank).
 - a. Testing showed that it was important to put the ferrofluid deep into the tank rather than pouring it directly on the surface.
5. The tank was then sealed with a fitting piece of glass and fast-acting epoxy similar to initial tank assembly.

Lessons Learned:

- How to iterate on a design
- Designing with safety and redundancy in mind
- Deeper understanding of electromagnetic forces
- Splitting a large design into multiple interlocking subsegments
- How to design PCBs
- Innerworkings of electrical components such as a Teensy or Raspberry Pi
- Leadership
- Teamwork
- Scheduling

5. Testing and Evaluation

5.1 Overview

Testing and evaluation were conducted to verify that the Smart Micro-Particle Manipulation Chamber met its functional and design requirements. The system was evaluated as a whole, including the user interface, communication architecture, electromagnet control system, tank design, and overall housing structure. Testing was conducted at both the subsystem and integrated system levels. Electrical components such as the electromagnets, Raspberry Pi, Teensy microcontroller, and PCB control boards were tested to ensure reliable communication and accurate electromagnet actuation. Mechanical components such as the glass tank, ferrofluid medium, and 3D printed housing were examined for structural integrity, visibility, and ease of assembly. Full system testing focused on the device's ability to accurately reproduce user input to the array, maintain a clear visualization of ferrofluid behavior, and operate safely under continuous use.

5.2 Testing and Results

Testing of the glass and tank and ferrofluid solution revealed that the tank was able to successfully contain and allow for the ferrofluid to react as hoped with the electromagnets. In addition, while there is some unfortunately inevitable minor staining through the usage of the ferrofluid, it is very minor and only a slight inconvenience when performing experiments. Testing of the magnets was a success as well, with the magnets being able to guide the ferrofluid both vertically and horizontally, overcoming the effects of gravity and complying with the program given to them. The magnets are able to perform their function without overheating both the tank and the housing unit, allowing for repeated testing without worry.

5.3 Assessment

The physical design of the chamber met requirement :[MESDP-015-MPMC-01], [MESDP-015-MPMC-05], [MESDP-015-MPMC-11], and [MESDP-015-MPMC-13]. The glass housing provided clear visibility of the ferrofluid movement while containing it. Ventilation slits were incorporated into the design to disperse heat buildup from extended operation of the electromagnets. The housing was designed with modularity and maintenance in mind as the walls, top, bottom, and electromagnet holder are separate units.

The system successfully controlled magnetic fields as specified in [MESDP-015-MPMC-02] and [MESDP-015-MPMC-04]. The electromagnet array is controllable by user inputs, with parameters that can be altered to suit the users needs.

The tablet interface satisfied requirements [MESDP-015-MPMC-08] and [MESDP-015-MPMC-12], providing an intuitive interface for interacting with the system. Users are able to easily generate patterns with visual representations provided in the app and simple touch controls.

Maintainability and cost requirements [MESDP-015-MPMC-18] and [MESDP-015-MPMC-19] were met. The inclusion of a bill of materials provided all the necessary information for mechanical, electrical, and software components to support easy replacements. The project remained within our budget constraints with minor additional funding requested for unforeseen component costs.

5.4 Future Plans

While the current prototype successfully demonstrates ferrofluid manipulation, many improvements can still be made to enhance or add system functionality. From a mechanical perspective, improvements can be made to the tank design to allow for ease of cleaning and fluid replacement. Modifications can also be implemented to improve ferrofluid buoyancy in the medium. Adjustments to the housing, such as more robust materials or a more compact design, could improve durability and thermal performance during extended usage. User interface enhancements could include drag-based drawing and pattern saving. Additionally, the communication system could continue to be developed to reduce latency, and improve overall system responsiveness. Future iterations can also explore more advanced control algorithms to allow for dynamic animation and more complex pattern generation. These improvements would increase the system’s functionality, reliability, and effectiveness as both a research and educational tool.

6. Schedule and Finances

Throughout the semester a Gantt chart has been utilized by the team in order to effectively stay on track and ensure that all work is being completed on time and equally. A chart like this clearly shows when due dates are, who is involved, and is shown below in Figure 5 and Figure 6.

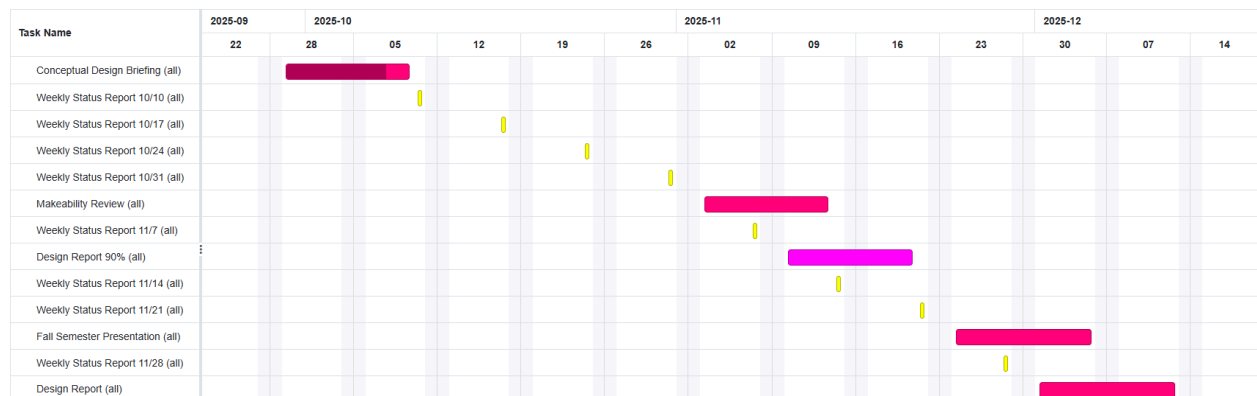


Figure 5: Fall 2025 Gantt Chart

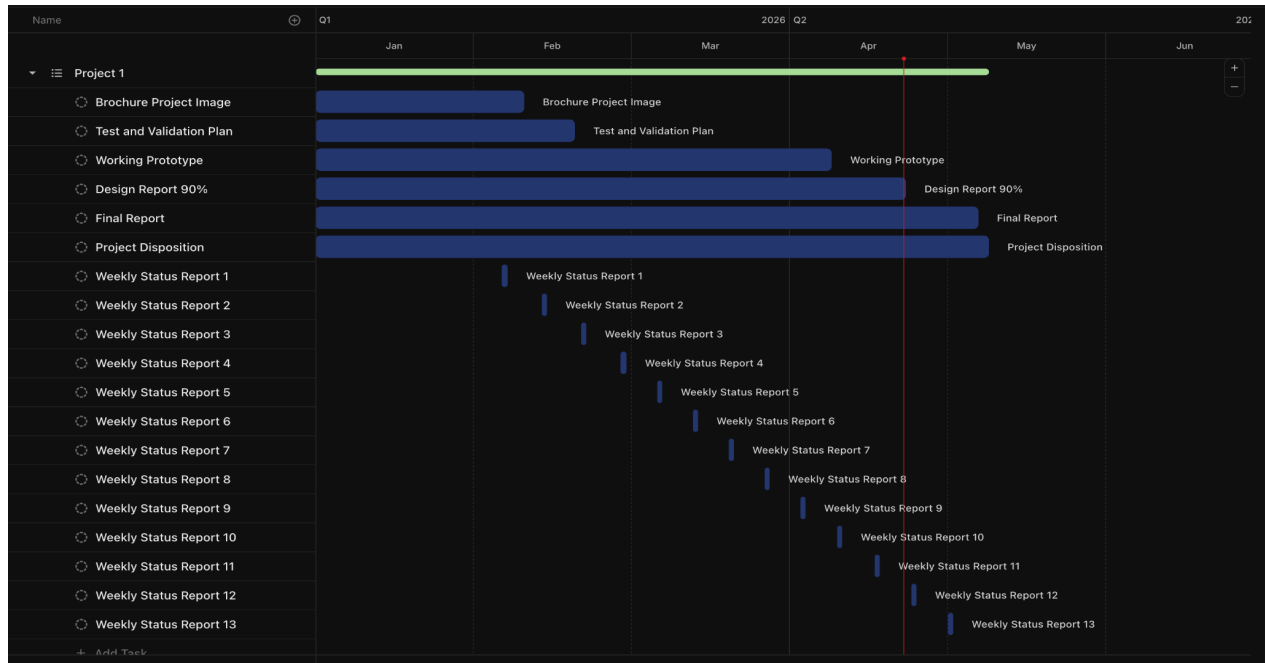


Figure 6: Spring 2026 Gantt Chart

The table below shows the current and accurate finance information for SDP115. While our group did go slightly over-budget, it was very marginal and necessary for the completion of the project. All parts were sourced through means that not only fit the budget, but allowed us to give the best product that we could by the end of the year.

Table One. Budget Tracker for Each Group

Mechanical Part Number	Part Name	Vendor	Quantity	Price	Links	Current Inventory	Projected Cost	Actual Spent
1	Glass Tank	Custom	1	~\$50	-	0	\$50	-
2	Screws	Lowes	2	\$2.48/8 Screws	M5 Screws	0	\$4.96	-
3	Nuts	Lowes	2	\$2.28/10 Nuts	M5 Nuts	0	\$4.56	-
4	Ferrofluid	Amazon	3 (60 mL each)	\$22.99	Ferrofluid	0	\$68.97	\$68.97
5	Housing	-	1	-	-	-	-	-
Electrical Part Number	Part Name	Vendor	Quantity	Unit Price	Links	Current Inventory	Projected Cost	Actual Spent

1	AC-DC Power Supply	Amazon	1	\$24.41	12V, 29A Power Supply	1	\$24.41	\$24.41
2	Assembled PCBs with ICs	JLCPCB	10	\$181.73	JLCPCB	0	~\$40	\$181.73
3	Electromagnets	Alibaba	100	\$4.00	JSP-1515	111	\$530.40	\$530.54
4	24 AWG Wire (10 colors)	Amazon	100 ft	\$41.98	24 AWG Wire	100 ft	\$41.98	\$41.98
5	2-pin connectors kit	Amazon	2	\$6.00	JST XH 2-Pin Connector Kit	2	\$12.00	\$12.00
6	2.54mm Header Strip	Amazon	800	\$5.49	HiLetgo Pin Headers	0	\$5.49	\$5.49
7	Set of 5 4-pin Dupont wires	Amazon	2	\$8.00	2.54mm Wire Female to Female	0	\$16	\$16
8	JST VH 2 Pin Silicone Cable	Amazon	1	\$9.69	2 Pin 22AWG Silicone Cable		\$9.69	\$9.69
9	M3.5 Fork Spade Crimp Connector	Home Depot	1	\$0.30	-		\$0.30	\$0.30
10	Amass XT60 Connector	Amazon	1	\$9.69	XT60 connector		\$9.69	\$9.69
11	XT60 Cable	Amazon	1	\$7.99	XT60 cable		\$7.99	\$7.99
12	Pre-crimped JST-VH cables	Amazon	1	\$24.99	JST-VH connector kit		\$24.99	\$24.99
13	DC DC CONVERTER 5V 5W	Digikey	2	\$12.93	TSR 1-2450		\$25.86	\$25.86
Software Part Number	Product Name	Vendor	Quantity	Price	Link	Current Inventory	Projected Cost	Actual Spent
1	Teensy 4.1 Microcontroller	Amazon	2	\$31.64	Teensy	0	\$63.28	\$63.28
2	Micro-USB Data Cable	Alibaba	1	\$0.12	Cable	0	\$0.12	\$0.12
3	Raspberry Pi 5	-	1	\$0.00	-	1	-	
4	Tablet	-	1	\$0.00	-	1	-	
5	ESP32 Camera	-	1	\$0.00	-	1	-	
FUNDING LIMIT	ACTUAL SPENT	REMAINING FUNDS						
\$950	\$1,023	-\$73						

7. Social Impact

The Smart Micro-Particle Manipulation Chamber is a research instrument with valuable educational and professional applications across a wide range of audiences. The device allows one to view electromagnetic interactions that were previously only observable at microscopic levels phenomena to be seen on the macroscopic scale, enabling students and educators with different technical backgrounds to visually understand and experience the concepts traditionally only found in high-technology research settings. The tablet-based interface requires no expertise or special training requirements to use, reducing the barrier to meaningful interaction with the system, and making it appropriate to be used in classroom demonstrations, lab courses, and even in outreach to the public in science. This type of technology has direct applications to a number of ongoing biomedical and materials research topics, such as targeted drug delivery, lab-on-a-chip diagnostics, and isolating microparticles. It is crucial that instrumentation that allows consistent, repeatable viewing of microfluidic behaviour exists and the development of these instruments eliminates the need to use expensive laboratory apparatus. In this sense, the chamber serves not just as a demonstration apparatus but as a valid low-cost research facility which can serve to enable experimental investigations at institutions where special equipment might not be readily available.

The modular housing structure, extensive documentation, and published bill of materials mean that the device can continue to operate and be modified long after the completion of the current project process. These individual components can be swapped in and swapped out, or upgraded without necessarily having to disassemble the entire system, and subsequent development teams can build upon the existing architecture without needing to write new code. This continuity in design makes the chamber an enduring platform and not a deliverable, adding to its educational and research value over time. On a larger scale, this project shows that it is possible to manufacture competent and well-designed scientific instrumentation using a student project budget. Since access to advanced research tools is still not evenly distributed among institutions, reproducible and cost-effective designs of this kind can make a positive contribution to a fairer research landscape - one where the right to engage in rigorous scientific research is no longer dictated by the funding available.

8. References

- [1] Cheng, Kunxue, et al. "Active Microparticle Manipulation: Recent Advances." *Sensors and Actuators A: Physical*, vol. 322, May 2021, p. 112616, <https://doi.org/10.1016/j.sna.2021.112616>. Accessed 19 Oct. 2022.
- [2] Wu, Jaun, et al. 2nd ed., vol. 25, IEEE, 2020, pp. 638–649, *Electrophoresis-Based Adaptive Manipulation of Nanowires in Fluid Suspension*.
- [3] "Ferrofluid Displays, Art, and Sculptures | Concept Zero," *Ferrofluid Displays, Art, and Sculptures | Concept Zero*. <https://www.czferro.com/>

[4] Minuti, Anca Emanuela, et al. "Fe-Cr-Nb-B Ferrofluid for Biomedical Applications." *Nanomaterials*, vol. 12, no. 9, Apr. 2022, p. 1488, <https://doi.org/10.3390/nano12091488>. Accessed 19 Sept. 2022.

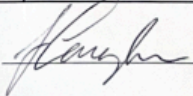
Appendix A - Design Makeability Review

Design Makeability Review

Design Makeability Checklist:

Project ID: **ME SDP 115**

		Not passed	Passed	N/A
Manufacturing process	Drawings		X	
	Dimensioning		X	
	Tolerances		X	
	Assembly drawings		X	
	Safety aspects of the design		X	
Bill of Materials	Parts		X	
	Materials		X	
	Cost		X	
	Vendors		X	
Software	Language, libraries		X	
	Development environment		X	
	Data flow diagrams		X	
Electrical	Circuit schematics		X	
	Circuit layouts		X	
	Proto board plans			X
	PCB		X	
	Device wiring		X	
	Color codes		X	
	Connections/terminals		X	

Reviewed by: 

Signed: _____

Date: 11/11/25

Design Makeability Review

Comments:

JST-XH

- Review data sheet for ¹2-pin connectors to ensure the connectors can support the amount of current drawn from the magnets
- Consider choosing an AC-DC Power Supply that can supply more than 20 A of current in case the magnets draw more current than expected. Highly suggested running tests to measure the amount of current the electromagnets draw.
- Order parts as soon as possible to avoid delays, especially parts from Alibaba.
- Ensure there are supports built in ^{the back wall} to ensure PCBs are well-contained
- Perform thermoanalysis on powered-on magnets to ensure safety of the magnet holder.
- Create a Gantt chart that ideally covers the full duration of the project through completion
- No dimensions should be over 14 inches
- System should use no more than 80% of the maximum supply current.

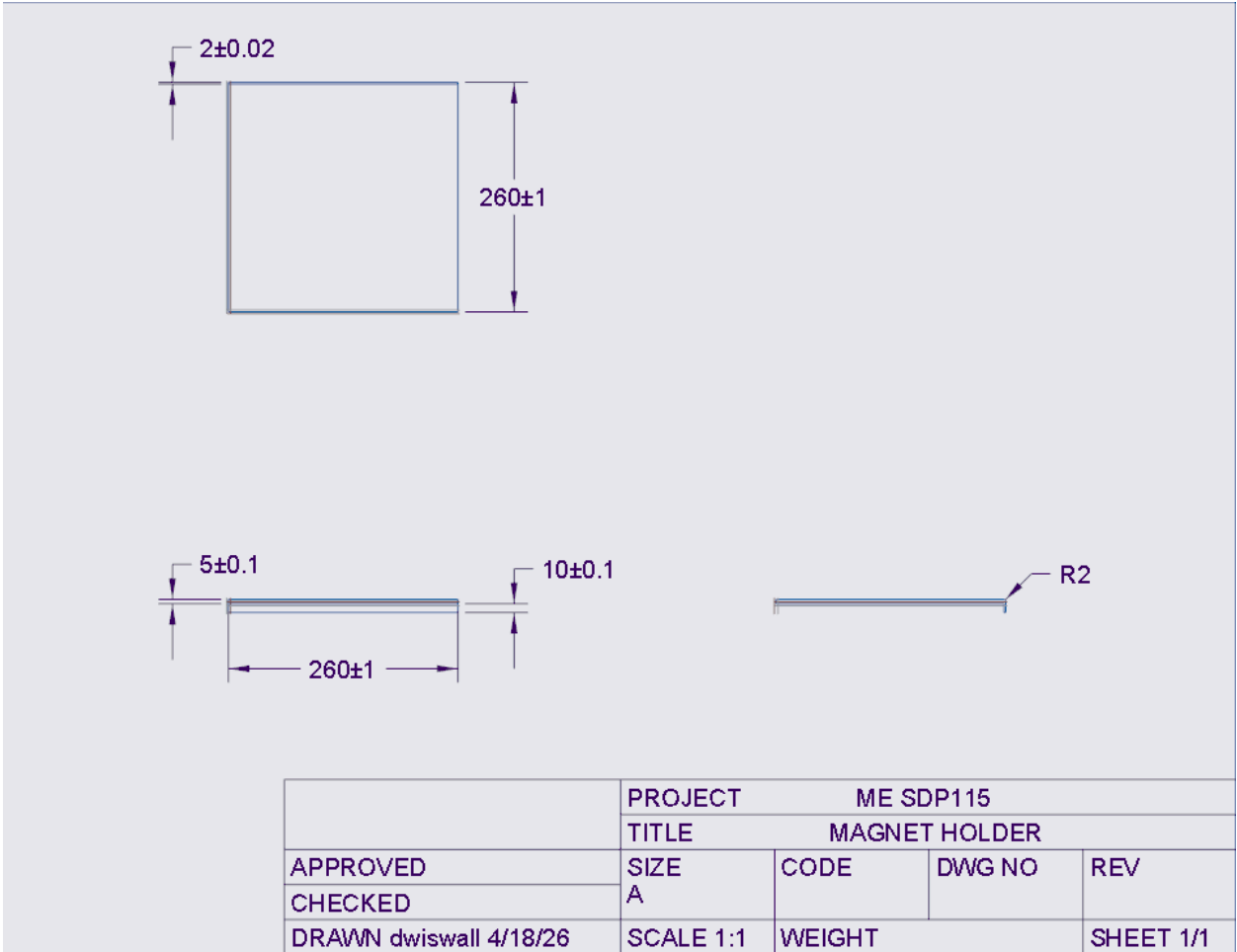
Appendix B - Bill of Materials

Mechanical Part Number	Part Name	Vendor	Quantity	Price	Links	Current Inventory	Projected Cost	Actual Spent
1	Glass Tank	Custom	1	~\$50	-	0	\$50	-
2	Screws	Lowes	2	\$2.48/8 Screws	M5 Screws	0	\$4.96	-
3	Nuts	Lowes	2	\$2.28/10 Nuts	M5 Nuts	0	\$4.56	-
4	Ferrofluid	Amazon	3 (60 mL each)	\$22.99	Ferrofluid	0	\$68.97	\$68.97
5	Housing	-	1	-	-	-	-	-
Electrical Part Number	Part Name	Vendor	Quantity	Unit Price	Links	Current Inventory	Projected Cost	Actual Spent
1	AC-DC Power Supply	Amazon	1	\$24.41	12V, 29A Power Supply	1	\$24.41	\$24.41
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5	2-pin connectors kit	Amazon	2	\$6.00	JST XH 2-Pin Connector Kit	2	\$12.00	\$12.00
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7	Set of 5 4-pin Dupont wires	Amazon	2	\$8.00	2.54mm Wire Female to Female	0	\$16	\$16
8	JST VH 2 Pin Silicone Cable	Amazon	1	\$9.69	2 Pin 22AWG Silicone Cable		\$9.69	\$9.69
9	M3.5 Fork Spade Crimp Connector	Home Depot	1	\$0.30	-		\$0.30	\$0.30
10	Amass XT60 Connector	Amazon	1	\$9.69	XT60 connector		\$9.69	\$9.69
11	XT60 Cable	Amazon	1	\$7.99	XT60 cable		\$7.99	\$7.99
12	Pre-crimped JST-VH cables	Amazon	1	\$24.99	JST-VH connector kit		\$24.99	\$24.99
13	DC DC CONVERTER 5V 5W	Digikey	2	\$12.93	TSR 1-2450		\$25.86	\$25.86
Software	Product Name	Vendor	Quantity	Price	Link	Current	Project	Actual

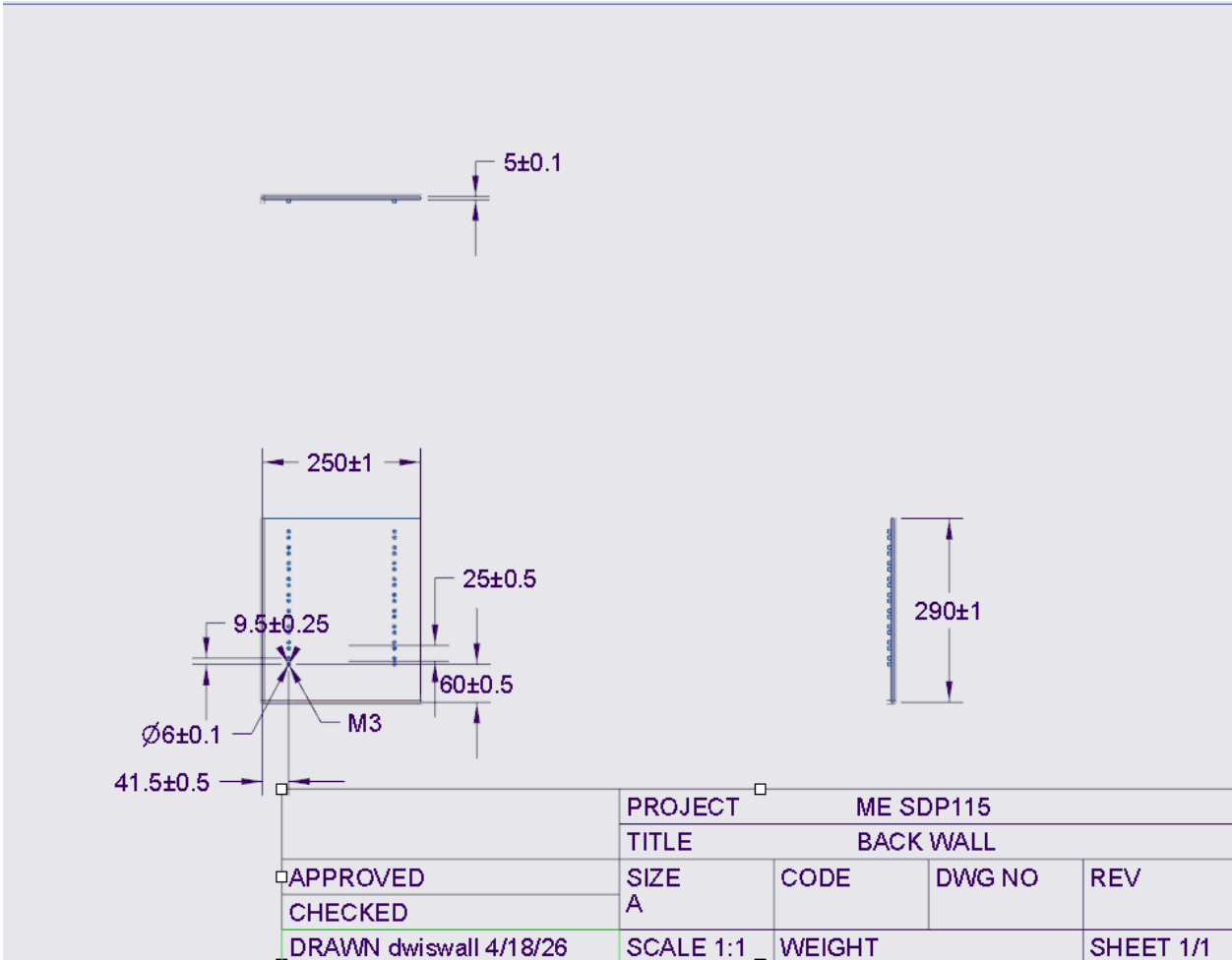
Part Number						Inventory	ed Cost	Spent
1	Teensy 4.1 Microcontroller	Amazon	2	\$31.64	Teensy	0	\$63.28	\$63.28
2	Micro-USB Data Cable	Alibaba	1	\$0.12	Cable	0	\$0.12	\$0.12
3	Raspberry Pi 5	-	1	\$0.00	-	1	-	
4	Tablet	-	1	\$0.00	-	1	-	
5	ESP32 Camera	-	1	\$0.00	-	1	-	

FUNDING LIMIT	ACTUAL SPENT	REMAINING FUNDS
\$950	\$1,023	-\$73

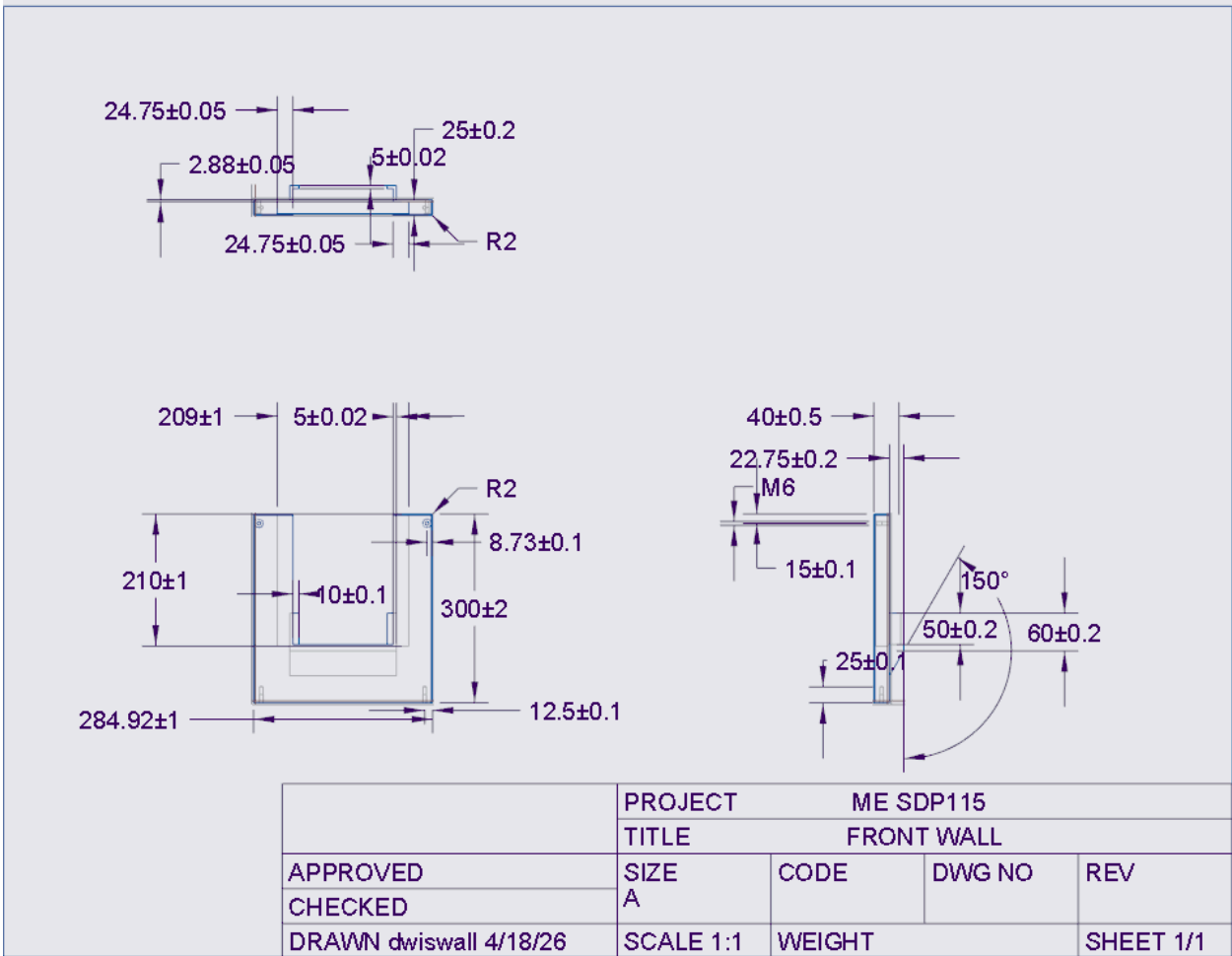
Appendix C: Detailed Drawings



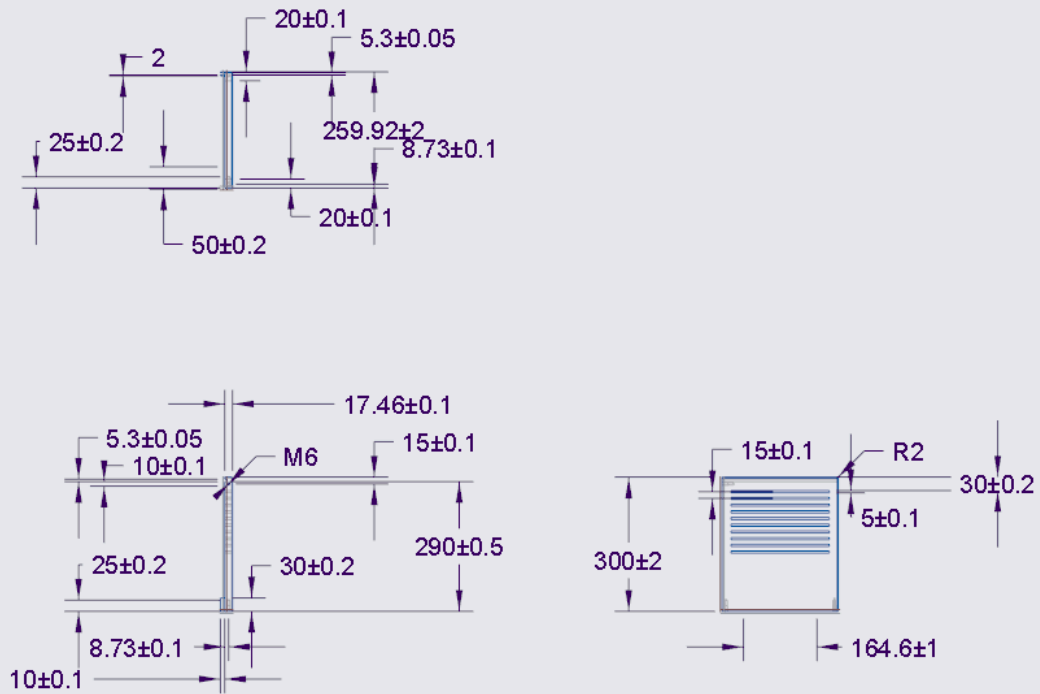
Drawing 1: Lid



Drawing 2: Back Wall

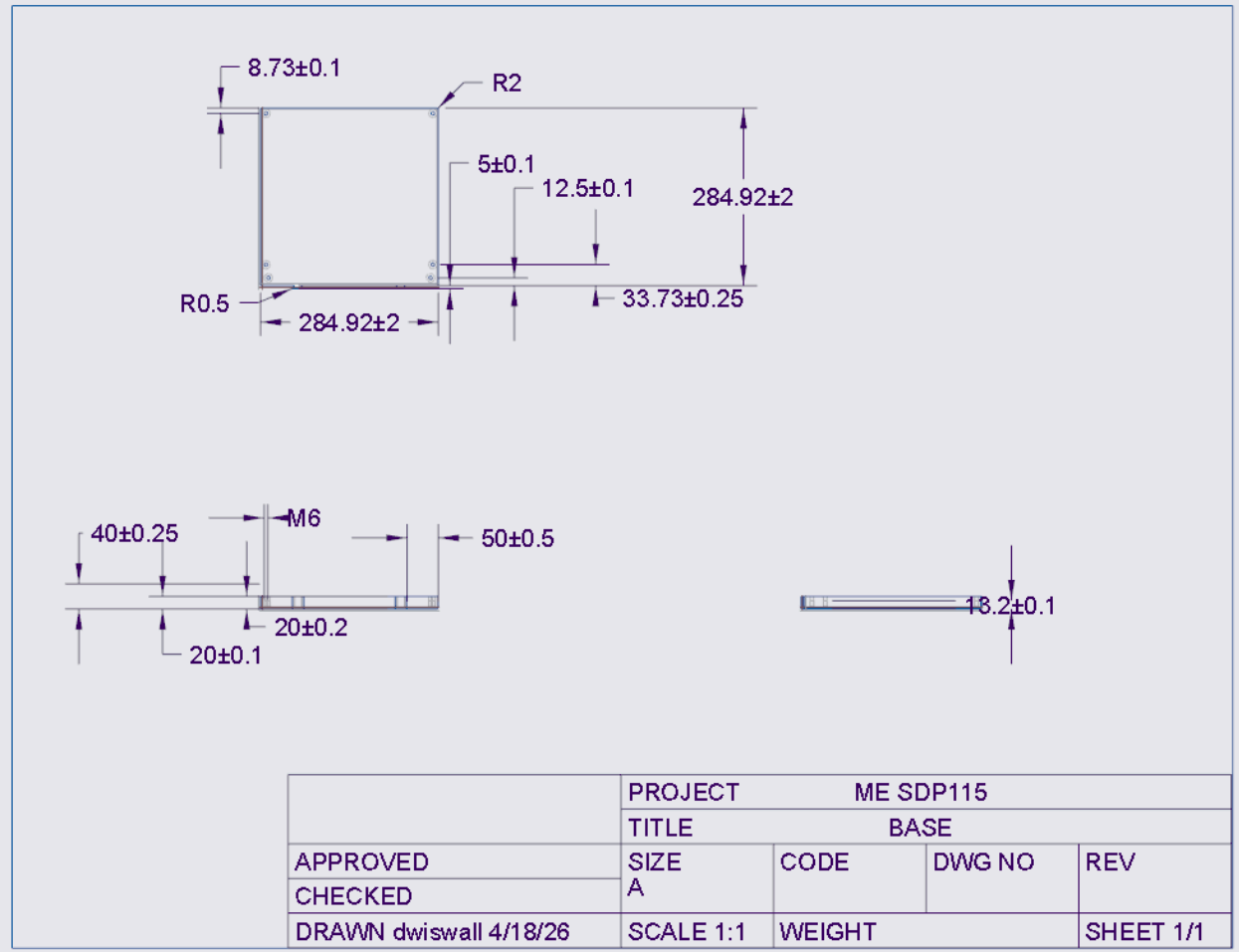


Drawing 3: Front Wall

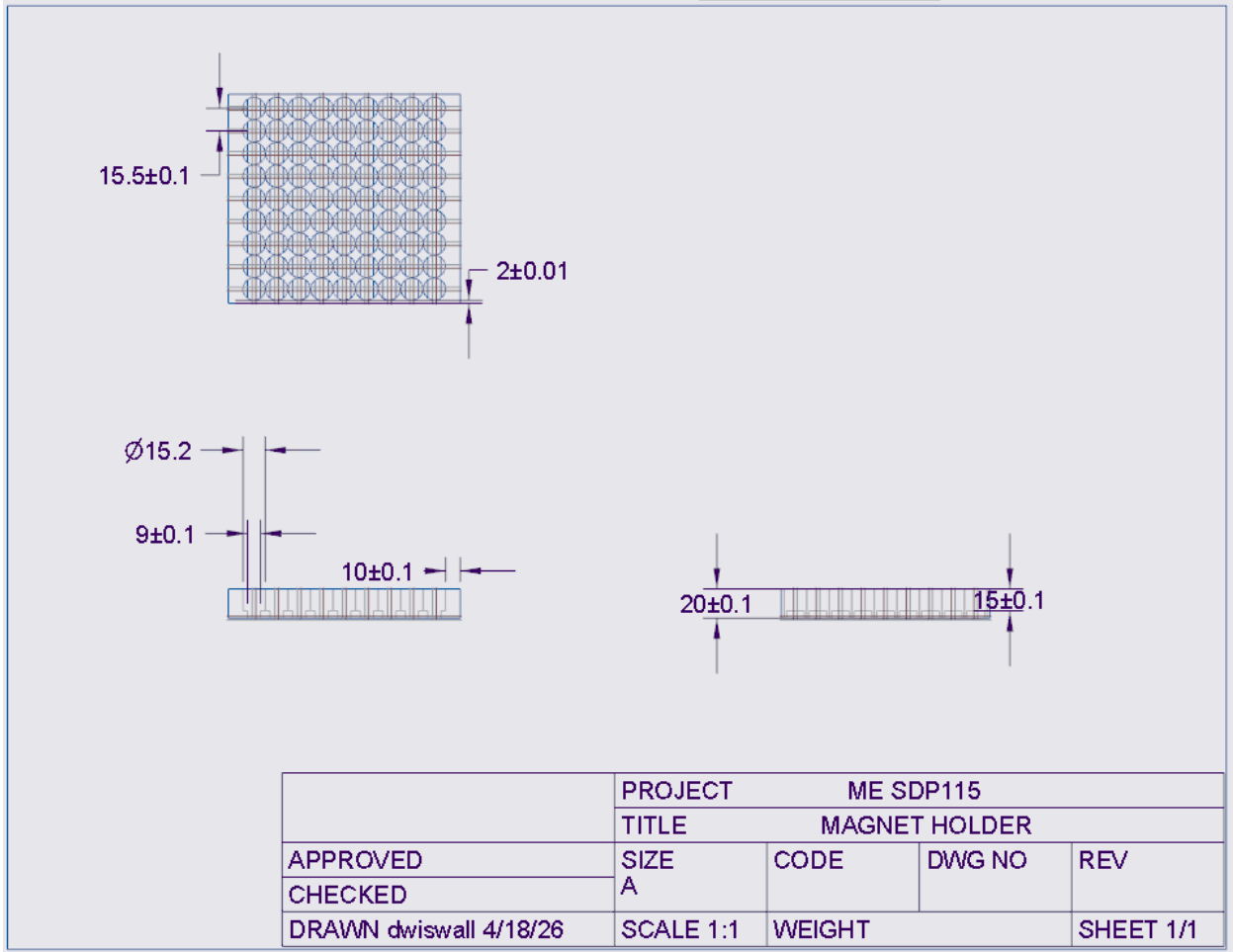


	PROJECT ME SDP115			
	TITLE SIDE WALL			
APPROVED	SIZE	CODE	DWG NO	REV
CHECKED	A			
DRAWN dwiswall 4/18/26	SCALE 1:1	WEIGHT		SHEET 1/1

Drawing 4: Side Wall



Drawing 5: Base



Drawing 6: Magnet Holder